# Sign test example

$X =$ length of time in sec between 2 calls to a call center

$H_0 : m = 6.2$

$H_1 : m < 6.2$

$\tilde{H}_1 : m > 6.2$

we will test against both alternatives

Data: 6, 7, 3, 2, 19, 10, 8, 4, 10, 7

signs of $X_i - m_0$: $\ominus$ $\oplus$ $\ominus$ $\ominus$ $\oplus$ $\oplus$ $\oplus$ $\ominus$ $\oplus$ $\oplus$

$S = \#$ of $\oplus$'s $= 6$ out of 10 $\sim \text{Binom}(10, \frac{1}{2})$

Significance level $\alpha = 10\%$.

ⓐ Testing $H_0: m = 6.2$ $\boxed{VS}$ $H_1: m < 6.2$,

We reject $H_0$ in favor of $H_1$

- if true median looks $< 6.2$
- if $P(X_i > 6.2) < P(X_i >$ true median$) = 0.5$
- if number of $\oplus$'s $(S)$ is too small

[as probability of each $\oplus$ is $< 0.5$]

Critical region $\boxed{C = \{S \le c\}}$ ($S$ is too small)

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Since Binom is discrete, it is not convenient to search for critical region explicitely

(there is no $c$: $P(S \le c) =$ exactly $0.1$)

p-value $= P(S \le \underset{\text{observed}}{6}) > 0.5 > 0.1 = \alpha \Rightarrow$ $\boxed{\text{accept } H_0}$

ⓑ Testing $H_0: m = 6.2$ $\boxed{VS}$ $\tilde{H}_1: m > 6.2$,

We reject $H_0$ in favor of $\tilde{H}_1$

- if true median $> 6.2$
- so, $P(X_i > 6.2) > P(X_i >$ true median$) = 0.5$
- so, $S$ is too big $\{S \ge c\} \to$ reject

p-value $= P\{S \ge 6\} = 1 - P(S < 6) =$

$= 1 - P(S \le 5) =$ (Table II)

$= 1 - 0.623 = 0.377 > 0.1$

p-value $> \alpha$

$\Rightarrow$ $\boxed{\text{accept } H_0}$